

Radiation-Hydrodynamics Simulation Verification Suite

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1 Setups for IMC and Rad-Hydro Simulations

This section contains exact physical benchmarks to verify supersonic radiative transfer implementations. The parameters are calculated to isolate local thermodynamic equilibrium (LTE) supersonic diffusion behavior against known analytical profiles.

1.1 Test 1 - Supersonic Henyey Wave (Power-Law Surface Drive)

For the first verification case, we deploy a supersonic Henyey wave matching a gold-like material model at a high-energy characteristic scale of $1 \text{ HeV} = 0.1 \text{ KeV} \approx 1.160452 \times 10^6 \text{ K}$. At this scale, the material energy density dominates, and independent radiation energy density ($a_R T^4$) can be neglected in the material energy balance during non-LTE tracking.

1.1.1 Material Model and Equation of State

The Rosseland mean opacity and specific internal energy are defined using power-law dependencies on temperature T under a constant spatially homogeneous material density ρ_0 . We set the material exponents to $\alpha = 1.5$ and $\gamma = 1.6$, with a density coupling parameter $\mu = 0$ and $\lambda = 0$.

The macroscopic opacity matches the parameterization:

$$k(T) = \rho_0 \kappa_R(T) = k_0 T^{-\alpha} = 10^4 \left(\frac{T}{\text{HeV}} \right)^{-1.5} \text{ cm}^{-1} \quad (1)$$

Thus, the opacity prefactor is $k_0 = 10^4 \text{ HeV}^{1.5}/\text{cm}$, which implies the face-averaging prefactor is:

$$g = \frac{1}{\rho_0 k_0} = \frac{1}{10^4 \rho_0} \text{ cm} \cdot \text{HeV}^{-1.5} \quad (2)$$

The specific internal energy equation of state follows:

$$e(T) = fT^\gamma = fT^{1.6} \quad (3)$$

By matching the total energy density to the standard benchmark equation $u(T) = \rho_0 e(T) = u_0 T^b$ (where $b = \gamma = 1.6$), we select $u_0 = 10^{13} \text{ erg}/(\text{cm}^3 \cdot \text{HeV}^{1.6})$. The material prefactor input is therefore:

$$f = \frac{u_0}{\rho_0} = \frac{10^{13}}{\rho_0} \text{ erg}/(\text{g} \cdot \text{HeV}^{1.6}) \quad (4)$$

1.1.2 Boundary Conditions and Driving Parameters

The problem imposes a power-law temporal surface temperature boundary condition at the slab edge ($x = 0$):

$$T_s(t) = T_0 t^s = 1.0 \times \left(\frac{t}{\text{ns}} \right)^{10/39} \text{ HeV} \quad (5)$$

where the temporal drive exponent is precisely the Henyey exponent $s = s_H$:

$$s_H = \frac{1}{4 + \alpha - b} = \frac{1}{4 + 1.5 - 1.6} = \frac{10}{39} \approx 0.256410 \quad (6)$$

For implementations running on an incident radiation field boundary condition (such as Implicit Monte Carlo or transport codes utilizing a Marshak/Milne boundary interface), the corresponding source bath temperature must be explicitly driven over time via:

$$T_{\text{bath}}(t) = T_s(t) [1 + \mathcal{B}]^{1/4} = \left(\frac{t}{\text{ns}} \right)^{10/39} \left[1 + 0.0576603 \left(\frac{t}{\text{ns}} \right)^{-8/13} \right]^{1/4} \text{ HeV} \quad (7)$$

This formulation yields an exact dimensionless boundary flux normalization value $S(0) = 2.003$ matching the internal solver metrics, which tracks the net energy flux flowing through the origin boundary.

1.1.3 Analytical Wavefront Verification Metrics

Because $s = s_H$, the similarity profile propagates at a perfectly constant wavefront velocity ($d = 1$), meaning the heat front position $x_F(t)$ changes linearly in time. Code developers should verify their spatial tracking arrays against the following baseline metrics at the final simulation time $t_{\text{final}} = 1 \text{ ns}$:

- **Heat Front Position:**

$$x_F(t) = 0.00118584 \times \left(\frac{t}{\text{ns}} \right) \text{ cm} \quad (8)$$

- **Total System Energy Integral:**

$$E_{\text{total}}(t) = 8.40870 \times 10^9 \times \left(\frac{t}{\text{ns}} \right)^{55/39} \text{ erg} \quad (9)$$

- **System Optical Thickness:** At $t = 1 \text{ ns}$, the cumulative optical depth across the heated domain scales to $\tau \approx 11.86$, proving the medium is sufficiently thick to validate the LTE diffusion equations against higher-order structural transport solvers.

1.2 Test 2 - Linear Front Wave (High-Temperature Regime)

[Content for Test 2 follows here...]